Traffic-Aware Task Reordering and Bi-Level Architecture for Lifelong Multi-Agent Path Finding

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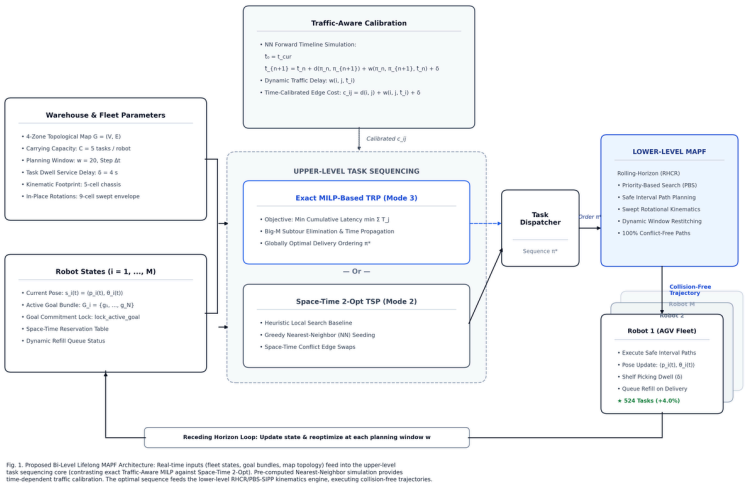
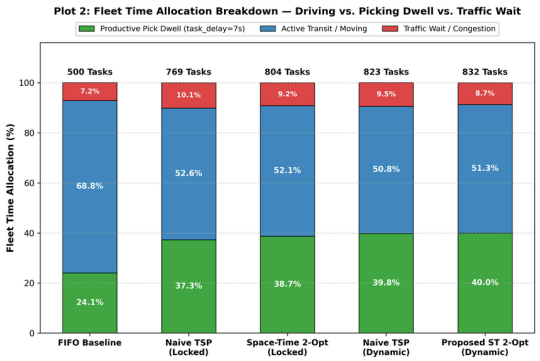


Fig. 1. Proposed Bi-Level Lifelong MAPF Architecture: Real-time inputs (fleet states, goal bundles, map topology) feed into the upper-level task sequencing core (contrasting exact Traffic-Aware MILP against Space-Time 2-Opt). Pre-computed Nearest-Neighbor simulation provides time-dependent traffic calibration. The optimal sequence feeds the lower-level RHCR/PBS-SIPP kinematics engine, executing collision-free trajectories.



(a) Proposed Bi-Level MAPF Architecture

(b) Fleet Time & Congestion Breakdown

1 Introduction/Motivation

2 Problem Statement & Objective

3 Methodology/Approach

4 Results & Findings

5 Conclusion & Impact

Demonstrates real-time industrial tractability and provable conflict elimination, enabling high-density autonomous fulfillment centers with zero deadlocks.

Keywords: Multi-Agent Path Finding | Rolling Horizon (RHCR) | Space-Time 2-Opt | MTZ-MILP | Warehouse Robotics